Synopsis

on

Civic Alert: A Road Hazard Intelligence System

Submitted in the partial fulfilment of the Degree of Bachelor of Technology (Information Technology)

Submitted by

Ustat Singh (Enrollment No: 00296303123)

Shivam Kalia (Enrollment No: 02196303123)

Under the supervision of

Dr. Minakshi Tomer

Department of Information Technology

Maharaja Surajmal Institute of Technology

Janakpuri, New Delhi.

2023-2027

Project Synopsis

Project Title

Civic Alert - AI/ML-Powered Road Hazard Detection and Management System

Abstract

Urban road infrastructure in India is plagued by persistent hazards such as potholes and waterlogging, which cause vehicular damage, traffic congestion, and serious accidents - yet civic authorities lack an efficient, real-time mechanism to detect, track, and resolve these issues. Civic Alert addresses this problem by proposing an AI/ML-powered, citizen-centric web platform that not only enables citizens to report road hazards but also ensures their complete resolution through a closed-loop system involving the Municipal Corporation of Delhi (MCD). The system enables citizens to upload geotagged images of road hazards via a React-based Progressive Web Application; these images are then analyzed in real-time by a YOLOv8 machine learning model - served through a FastAPI microservice - to automatically classify hazard types (pothole, waterlogging, etc.) with a configurable confidence threshold. Additionally, the platform leverages Generative AI powered by ChatGPT (OpenAI) for intelligent hazard analysis, contextual descriptions, and enhanced decision support. Once a hazard is validated and reported, the system automatically notifies the MCD through an administrative dashboard, where authorities can review the report and assign repair tasks to field workers. After the repair is completed, the assigned MCD worker is required to capture and upload a photograph of the fixed site; this "after-fix" image is then cross-verified by the AI/ML system against the original hazard image to confirm that the issue has been genuinely resolved before marking it as fixed. The platform incorporates Google OAuth 2.0 authentication to prevent spam, an interactive Leaflet-based map view for geospatial hazard visualization, a live feed of nearby alerts, and role-based dashboards for citizens, administrators, and workers. By combining crowdsourced citizen reporting with ML-based detection, Gen AI-powered analysis, and an AI-verified repair-confirmation loop, Civic Alert creates an end-to-end smart-city feedback system that accelerates hazard resolution, ensures accountability, and improves urban road safety.

Objectives

1. Automate Road Hazard Detection — YOLOv8-based hazard classification from images
2. Enable Real-Time Citizen Reporting — Mobile PWA with geotagged image uploads
3. Provide Geospatial Hazard Visualization — Interactive real-time map (Leaflet/OSM)
4. Implement End-to-End Hazard Lifecycle Management — Report → validation → assignment → resolution
5. Ensure Platform Integrity via Authentication — Google OAuth 2.0 + role-based access control
6. Facilitate AI/ML-Verified Resolution — Before/after repair verification via YOLOv8 + ChatGPT
7. Deliver Proactive Safety Alerts — Trip Mode with route-based audio notifications

Technologies Used

Layer	Technology	Purpose
Frontend	React 18, Vite 5, Tailwind CSS	SPA with component-based architecture and modern build tooling
Frontend Libraries	React Router DOM, Leaflet / React-Leaflet, Lucide React, Axios	Client-side routing, interactive map rendering, iconography, HTTP communication
PWA Support	Vite Plugin PWA	Installable progressive web app for mobile use
Backend	Java 21, Spring Boot 4.0	RESTful API server with MVC architecture, DI, and ORM support
Security	Spring Security, OAuth2 Client	Google OAuth 2.0 authentication and role-based authorization
Database	PostgreSQL	Persistent relational storage for users, hazard reports, and assignments
ORM	Spring Data JPA, Hibernate	Object-relational mapping and database abstraction
ML Model	YOLOv8	Real-time object detection and image classification for hazard identification
ML API Server	Python, FastAPI, Uvicorn	High-performance async REST API for serving ML model predictions

Generative AI	ChatGPT (OpenAI API)	Gen AI-powered intelligent hazard analysis, description generation, and repair verification
Image Processing	Pillow (PIL)	Image preprocessing and format conversion before model inference
Build Tools	Maven, npm	Dependency management and build automation for backend and frontend
Utility	Lombok, Jackson	Boilerplate code reduction and JSON serialization/deserialization

Team Members

Ustat Singh – 00296303123

Shivam Kalia – 02196303123

Preferred Mentor

Dr Minakshi Tomer

Literature Review

Pothole detection and road hazard management have become active areas of research in recent years, driven by advances in deep learning, computer vision, and IoT-based smart city initiatives. The table below summarizes eight recent and relevant studies that either apply ML/AI techniques to pothole detection,

propose citizen-centric reporting systems, or attempt to characterize road anomalies for maintenance prioritization.

Sr.	Paper / Source	What it did	Limitation / Gap left open
1	Kumari et al., "YOLOv8-Based Real-Time Pothole Detection with GPS Alerts and Civic Reporting" (TKR College, 2025)	Automated pothole detection using YOLOv8 with GPS geotagging, severity classification (Low/Medium/High), government dashboard sorted by severity, and real-time FCM/Socket.io notifications. Used MongoDB for storage.	No repair-verification mechanism. No AI cross-verification of whether pothole was fixed. Lacks citizen authentication to prevent spam reports.
2	Bhosale & Ponnusamy, "Smart Pothole Detection and Reporting System" (Sandip University, EDP Sciences, 2025)	Real-time detection using CV and ML with vehicle-mounted cameras. GPS for location tagging, automated alerts to PWD/RTO/municipal authorities, GIS-based interactive map dashboard for maintenance teams.	Limited to vehicle-mounted cameras, reducing citizen participation. No post-repair verification. No Gen AI integration. GPS accuracy issues in dense urban areas.
3	Baroudi et al., "Enhancing Pothole Detection: Integrated Segmentation and Depth Estimation" (KFUPM, Saudi Arabia, 2025)	YOLOv8-seg model for instance segmentation with monocular depth estimation via Kinect V2. Novel dataset with depth ground truth. Comprehensive characterization including area and depth for	Dashboard-camera input only. Dataset from Saudi Arabia may not generalize to Indian roads. No citizen reporting interface. Depth estimation needs specialized hardware (Kinect V2).
4	"POT-YOLO: Edge Segmentation-Based YOLOv8 Network" (IEEE Sensors Journal, Vol. 24, Aug 2024)	POT-YOLOv8 approach with edge segmentation for real-time detection. Applied CAGF for image preprocessing to handle variable lighting and surface textures, improving robustness.	Purely detection-focused. No reporting, management, or civic engagement component. No severity classification or repair tracking system.
5	"Pothole Detection using YOLOv8 and CNN" (IEEE INOCON 2024, March 2024)	Two-stage pipeline: CNN (ResNet50) for initial classification followed by YOLOv8 for precise localization. Explored image preprocessing methods for diverse conditions.	Adds computational overhead. No GPS integration, citizen reporting, municipal dashboard, or post-repair verification. Limited to accuracy benchmarking.
6	"Automated Pothole Detection using Transfer Learning" (IEEE I2CT, April 2024)	Transfer learning with YOLOv8 fine-tuned on custom datasets combined with drone imagery. Framework for road maintenance mapping from aerial and ground-level images.	Drone-based approach has high operational cost. Not suitable for citizen-driven reporting. No repair lifecycle management or resolution confirmation.

7	Murty et al., "Pothole Detection using CNNs" (2024)	Compared ResNetV2, ResNet50, VGG19, and YOLOv8 for pothole detection. Benchmarked detection accuracy to identify best-performing architecture.	Model comparison only - no system-level implementation. No geospatial visualization, severity classification, citizen engagement or municipal workflow.
8	Majidifrad et al., "YOLO + U-Net for Road Anomaly Detection and Severity" (2023)	Combined YOLO for detection with U-Net for semantic segmentation to identify anomalies and categorize severity simultaneously.	No full pothole characterization. No mobile/web citizen reporting. No municipal notification. No before-and-after image comparison for repair verification.

Research Gap Analysis

From the literature reviewed above, a clear pattern emerges. Existing studies primarily fall into two categories:

> **Detection-only studies:**

They focus on training and benchmarking ML models (YOLOv8, CNNs, R-CNNs) for pothole detection accuracy, but do not build a complete system around the model. They lack citizen-facing interfaces, municipal dashboards, authentication, notification services, and repair workflows.

> **Detection + Reporting studies:**

Some studies extend detection with GPS tagging and authority notification, but still stop short of a complete hazard lifecycle management system. Critically, none of the reviewed papers implement:

- AI/ML-verified repair confirmation (requiring MCD workers to upload after-fix images for cross-verification)
- Gen AI integration (using ChatGPT for intelligent hazard analysis and contextual decision support)
- Anti-spam authentication (OAuth 2.0 to ensure genuine, traceable reports)
- Proactive commuter safety alerts (Trip Mode with proximity-based audio warnings)

Civic Alert bridges this gap by combining YOLOv8 ML-based detection with ChatGPT-powered Gen AI analysis, a complete MCD notification and worker assignment system, mandatory after-fix image upload with AI/ML cross-verification, and role-based dashboards - creating the first end-to-end, AI/ML-powered pothole detection, reporting, and resolution-verification platform.